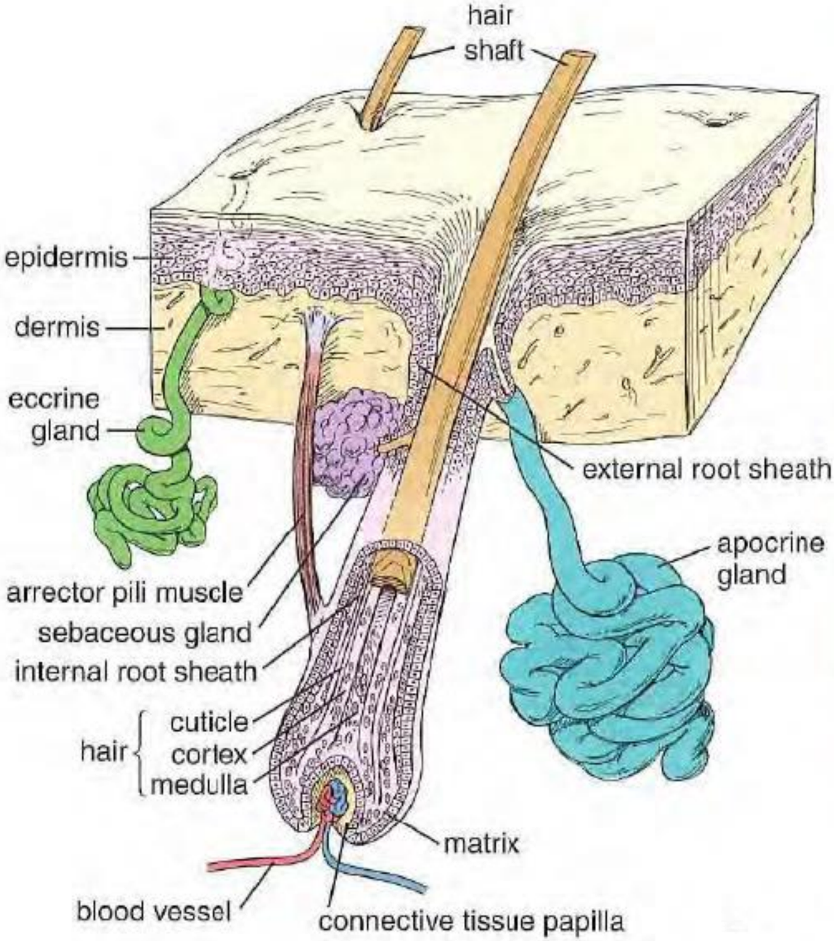
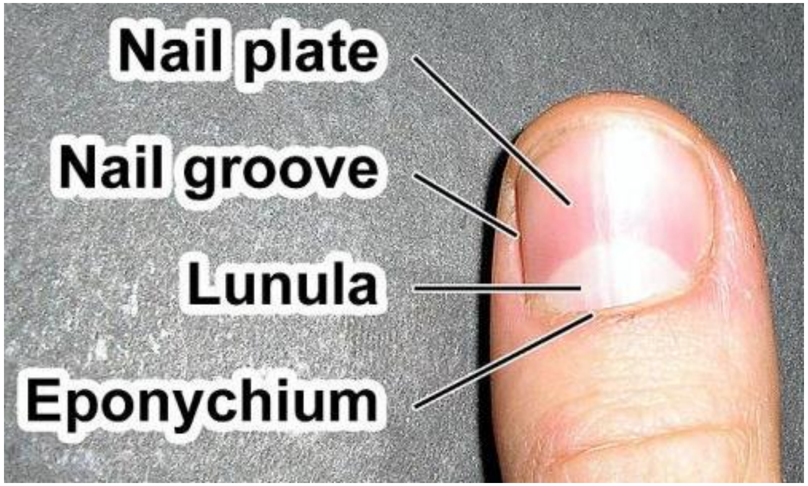
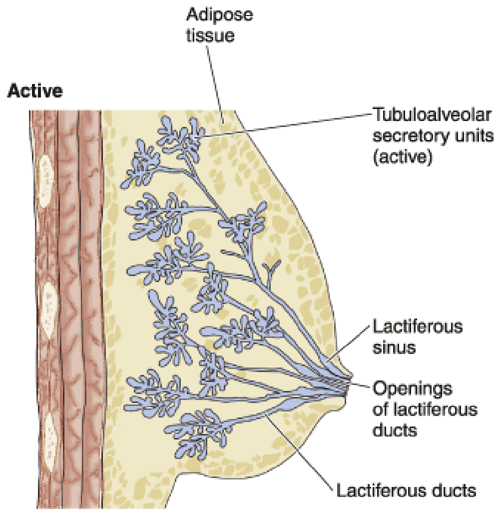
Integumentary System



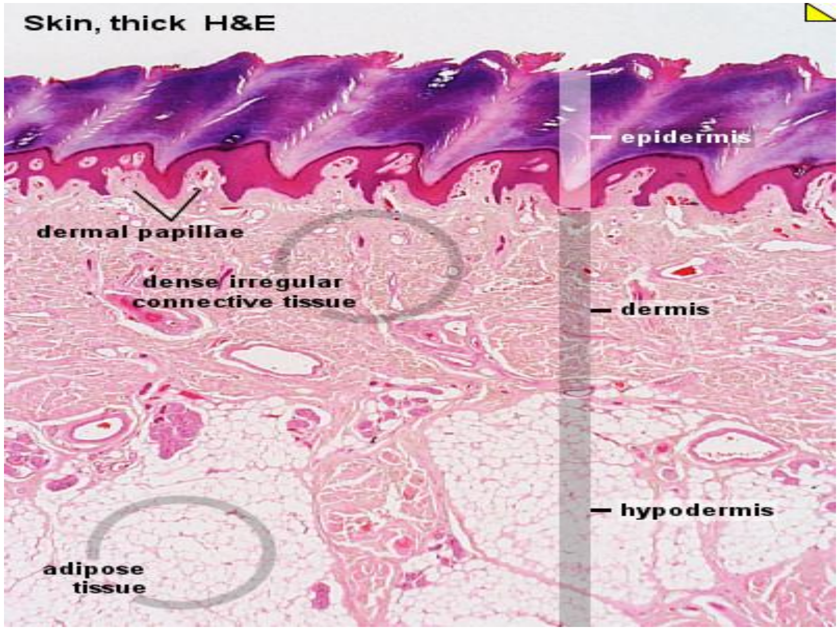




Fikre Bayu

Skin

- Has two layers:
  1. **Epidermis**
  2. **Dermis**
- 15-20% of body weight
- With 1.2-2.3 m^2 of surface area
- 0.5 mm (on eyelid) to 4 mm (on the back) in **total thickness** (both dermis & epidermis)



Functions Skin

- **Prevent** extreme **water loss**
- Act as **receptor** organ
- **Protect** the body from impact and friction injuries, and UV rays
- Provides **immunological** information to the lymphoid tissue.
- By its glands, blood vessels and adipose tissue participates in **thermoregulation, body metabolism** and **excretion of various substances**
- Has **endocrine** function as it produces **cytokines, growth hormones** and activates formation of **Vit D_3** that facilitates the absorption of Ca & PO_4
- Has **exocrine** function through its **sweat, sebaceous** and **apocrine glands**

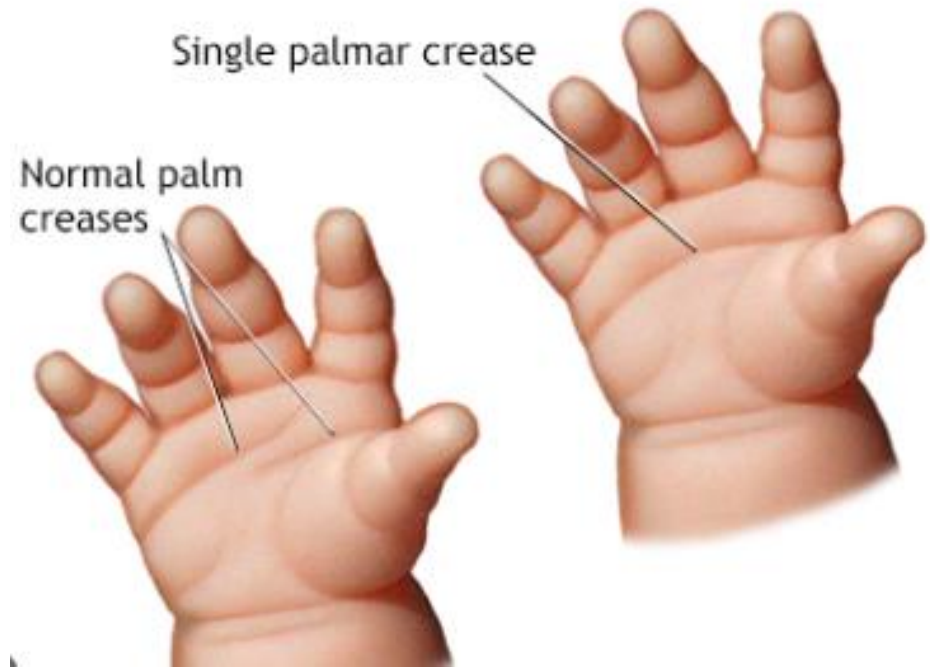
Skin

- Shows three types of markings:

1. **Tension lines**
   - **Network of fine shallow grooves** giving polygonal areas
   - By attachments of **collagen fibres in the dermis**

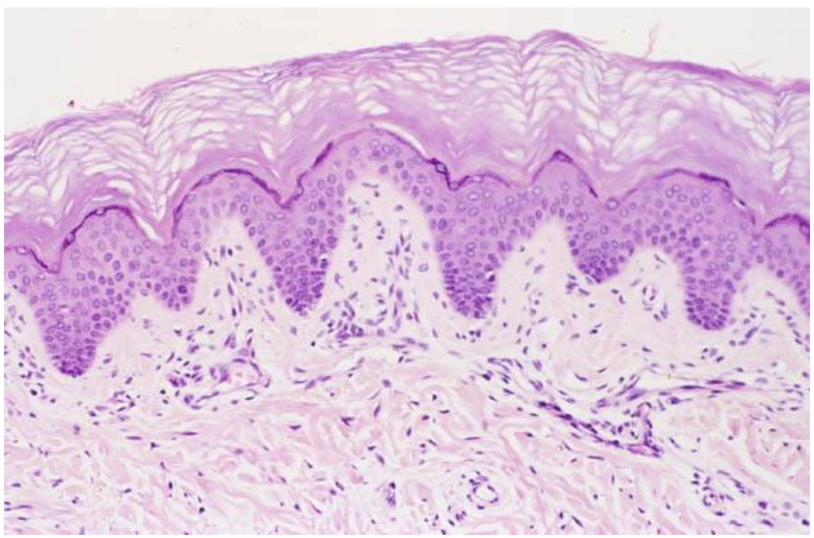
Skin

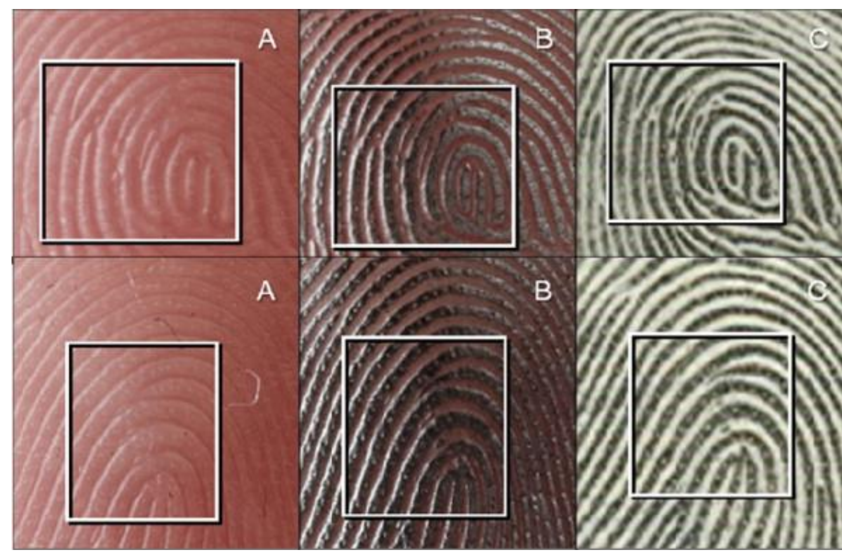
2. **Flexure lines**
   – Correlates to folds in dermis associated **with movements of the joints**
   – By attachments of **dermis to the underlying deep fascia**
   – Conspicuous on wrist, palms, soles, fingers and toes



Skin

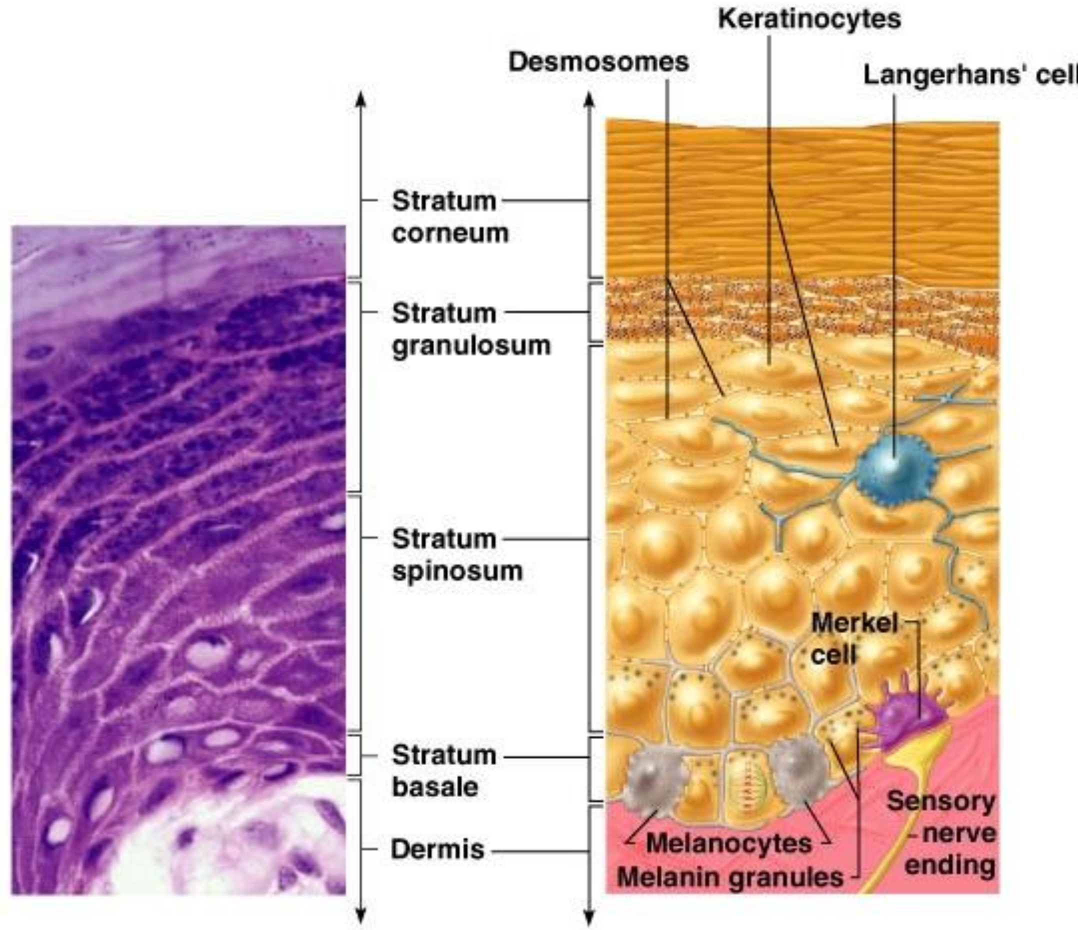
3. **Papillary ridges (finger prints)**
   - By underlying **dermal papillae** and **ridges**
   - As **loops, arches and whorls**





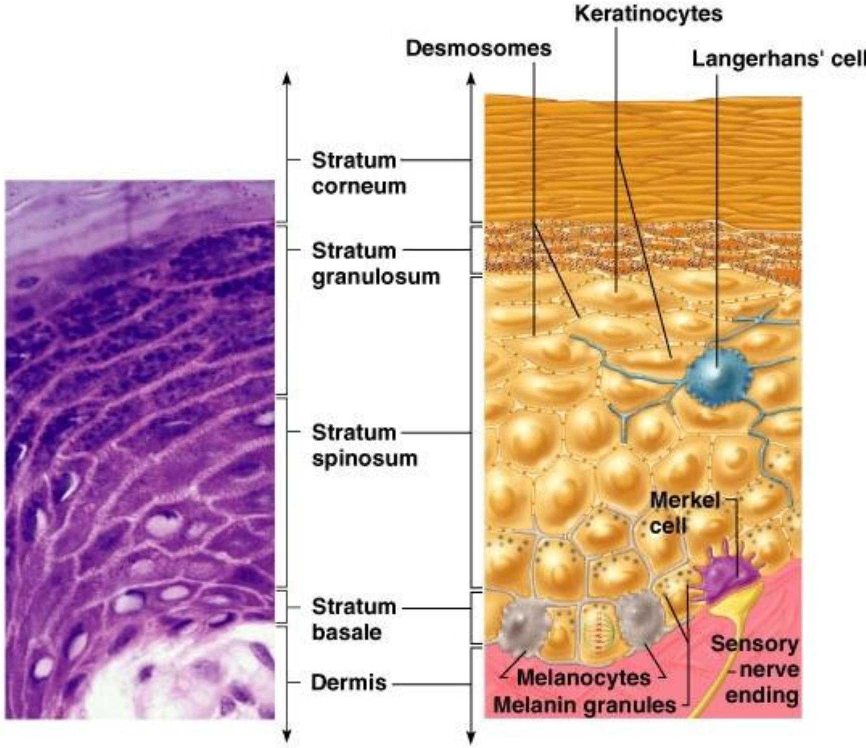
Epidermis

- Has four types of cells:
  1. **Keratinocytes**
     - **In five layers**
  2. **Melanocytes**
  3. **Langerhan's cells**
  4. **Merkels's cells**



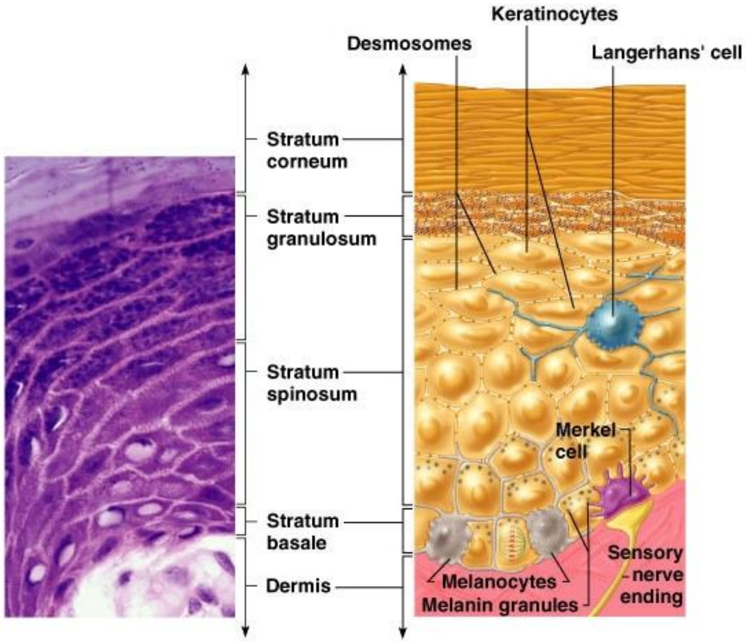
Stratum basale (Stratum germinativum)

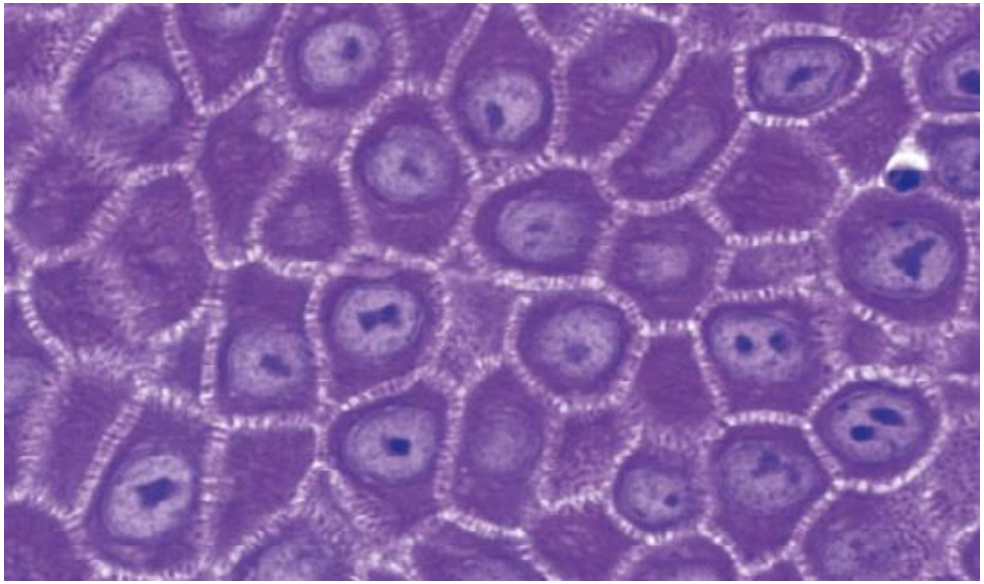
- By a single layer of **basophilic low columnar cells** on basal lamina.
- Contains many **intermediate filaments that will mature as a component of keratin**
- Large number of **desmosomes & hemidesmosomes**



Stratum spinosum (Prickle cells layer)

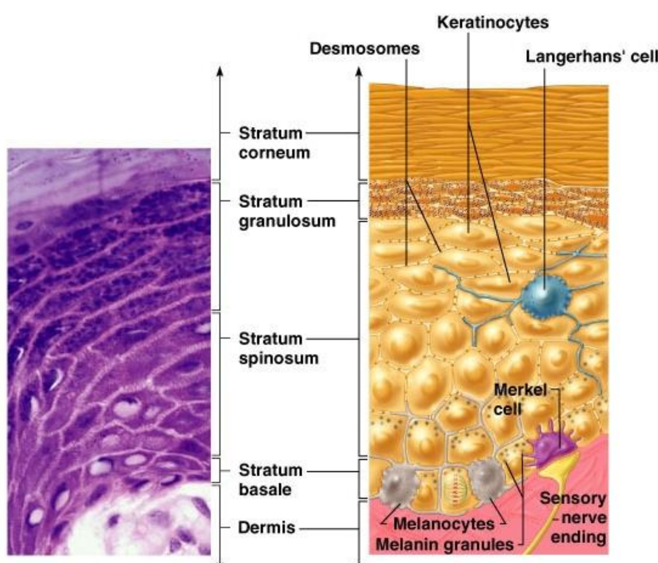
- Several layers of **polygonal, cuboidal or slightly flattened cells**
- Deeper cells same as stratum basale show mitotic activity and give **Malpighian layer.**
- Firmly bound by **desmosomes** that give a **spinney appearance**

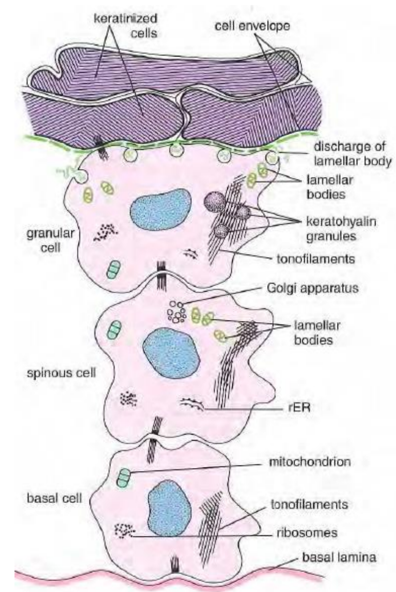




Stratum granulosum (Granular cells layer)

- 3-5 layers of polygonal and flattened cells
- Cells die with **breakdown of the nucleus** and **organelles** and **thickening of the plasma membrane**
- Cells contain **2 types of granules**
  1. **Non-membrane bound Keratohyaline granules -** for formation of **keratin with the intermediate filaments**
  2. **Membrane coated lamellar granules -** form **lipid enevelop** of **epidermal water barrier**





Stratum granulosum (Granular cells layer)

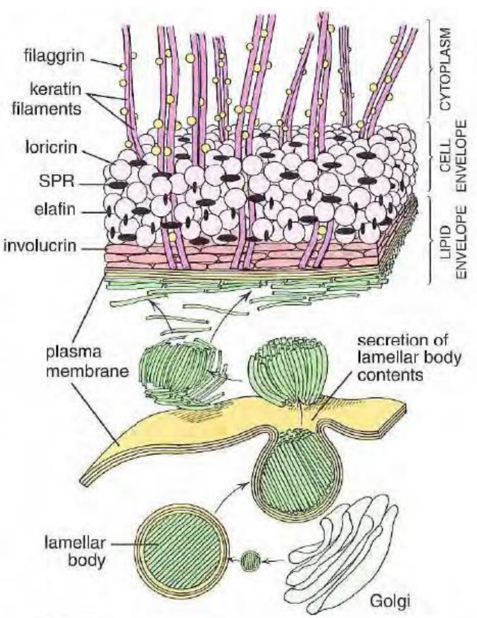
Epidermal water barrier

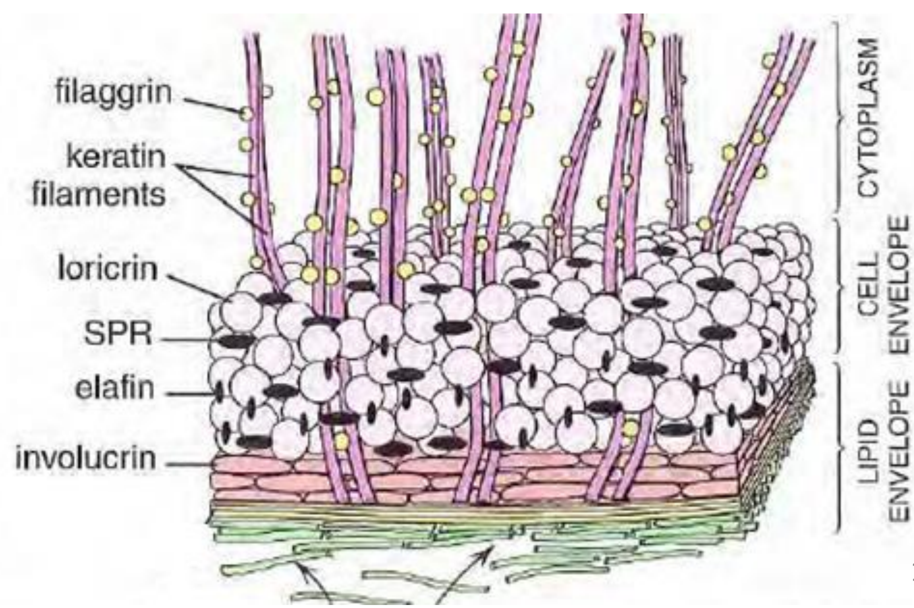
a. **Lipid envelope**

– By various types of **lipids** bound to the external surface of the plasma membrane

b. **Cell envelope**

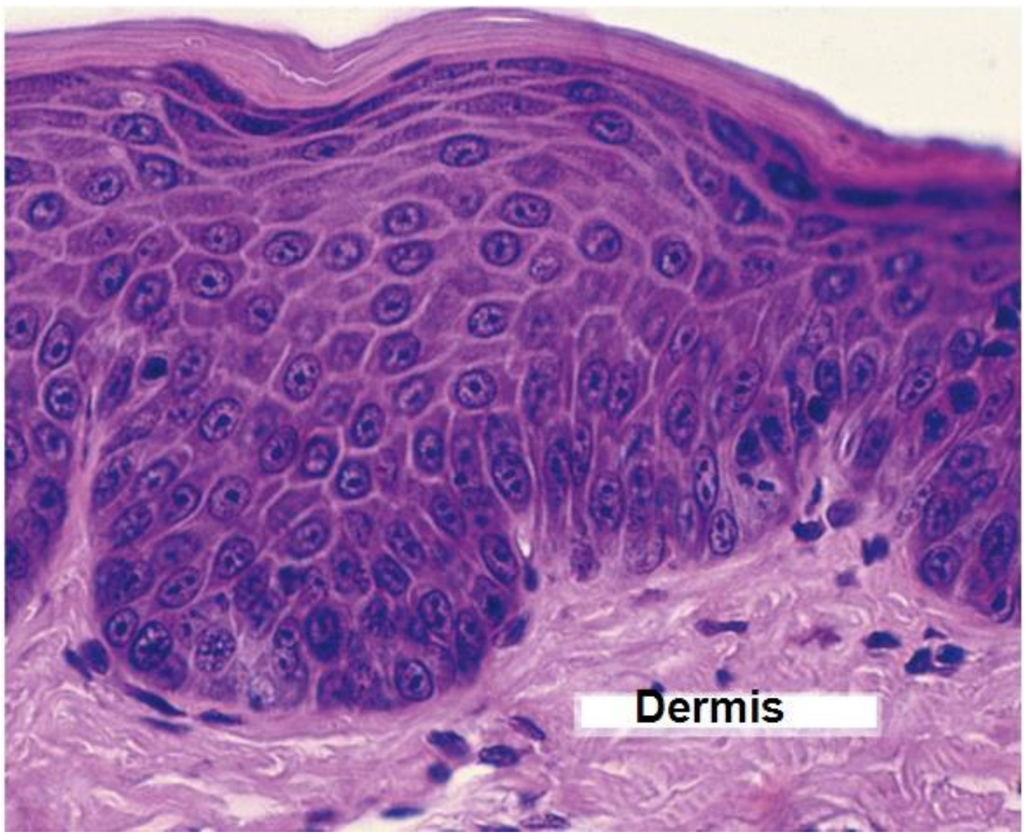
– By various types of **proteins**

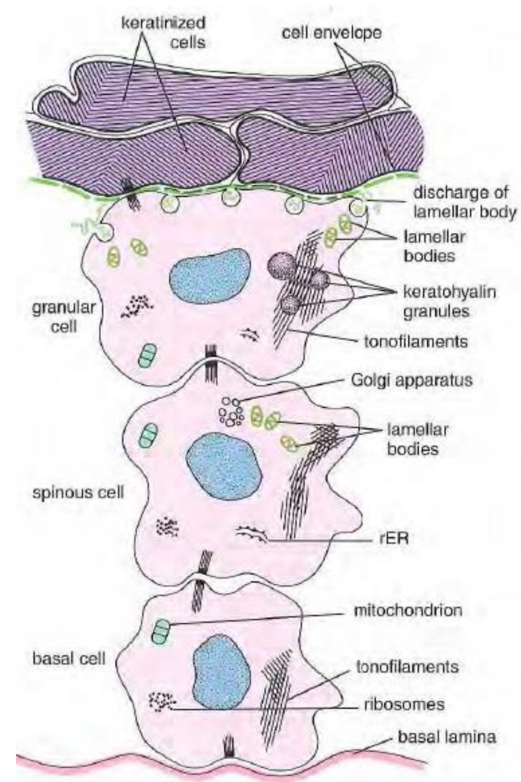




Stratum lucidum

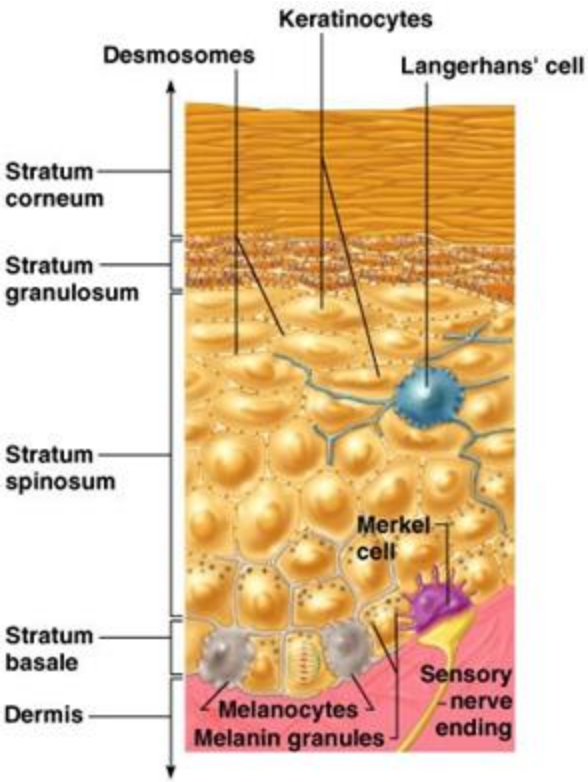
- May **be absent in thin skin**
- Flattened **dead cells** with no nuclei and organelles
- Densely packed filaments embedded in an electron-dense matrix

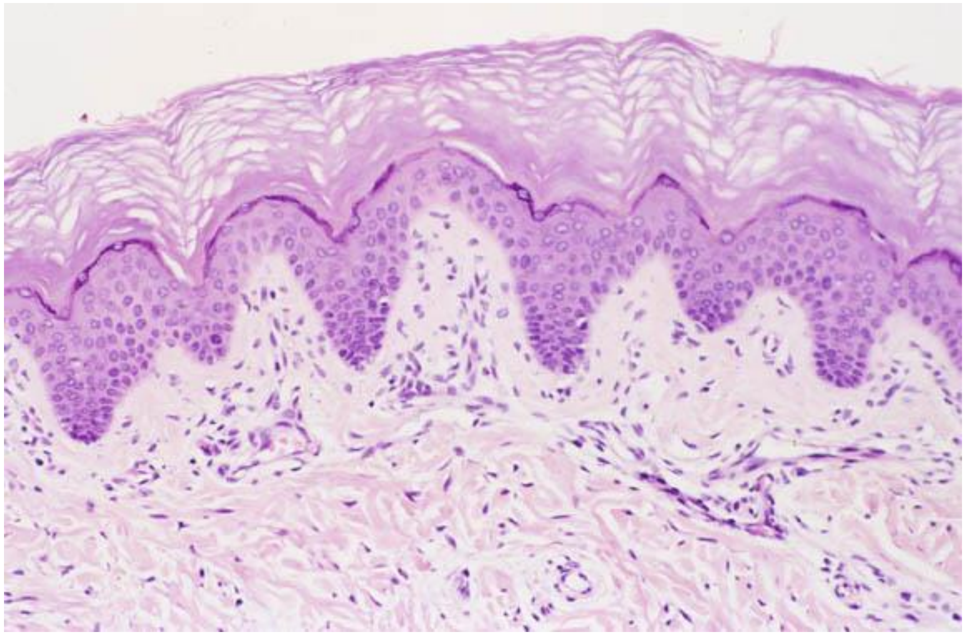




Stratum corneum

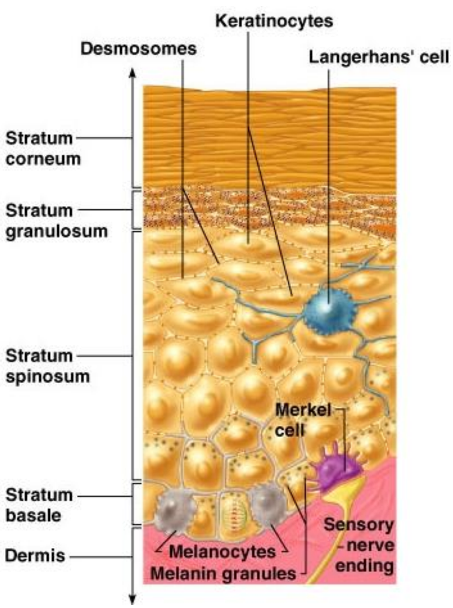
- Several layers of **flattened dead cells**
- **Thick plasma membrane** filled with **tonofilaments embedded in amorphous matrix (keratin)**

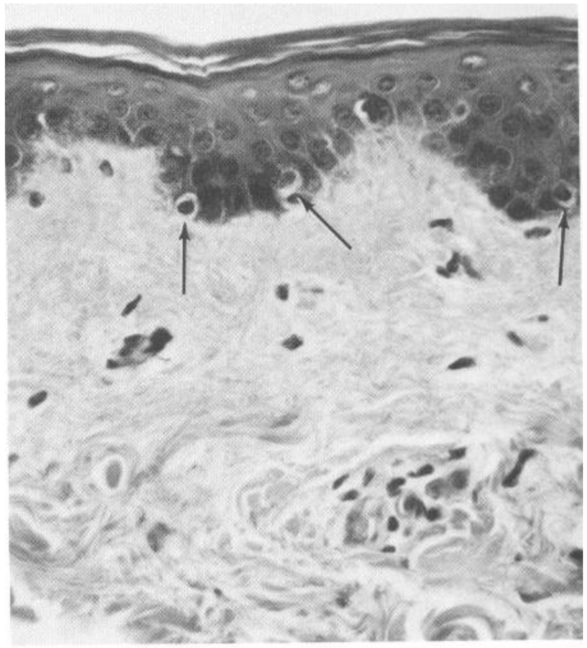


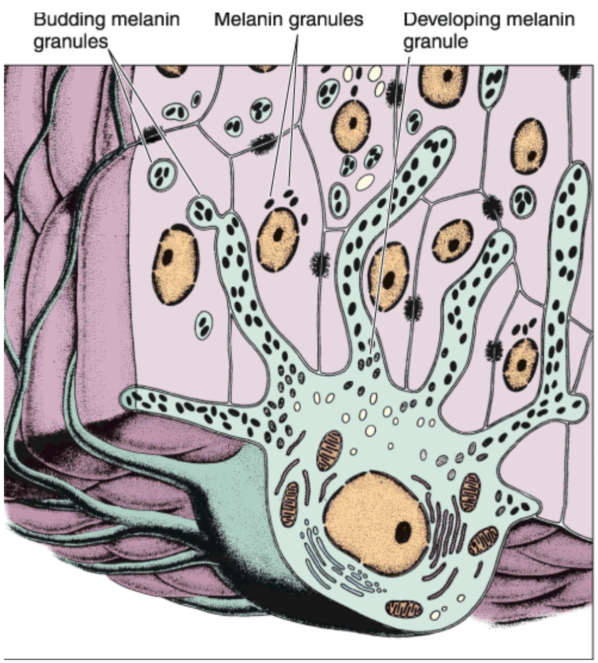


Melanocytes

- Located **beneath** or **between cells of stratum basale**
- Also called **dendritic cells**
- **Have a ratio of 1:4 to 1:10** and is constant in all races
- With **elongated nucleus** surrounded by a **clear cytoplasm and long branched processes**







Colour of skin

- Depends on **several factors,** such as:

1. **Rate of melanin synthesis** and **degradation**
2. **Form of melanin present,** as it exists in **two forms**:
    a. **Eumelanin – a brownish black pigment**
    b. **Pheomelanin – a reddish yellow pigment**

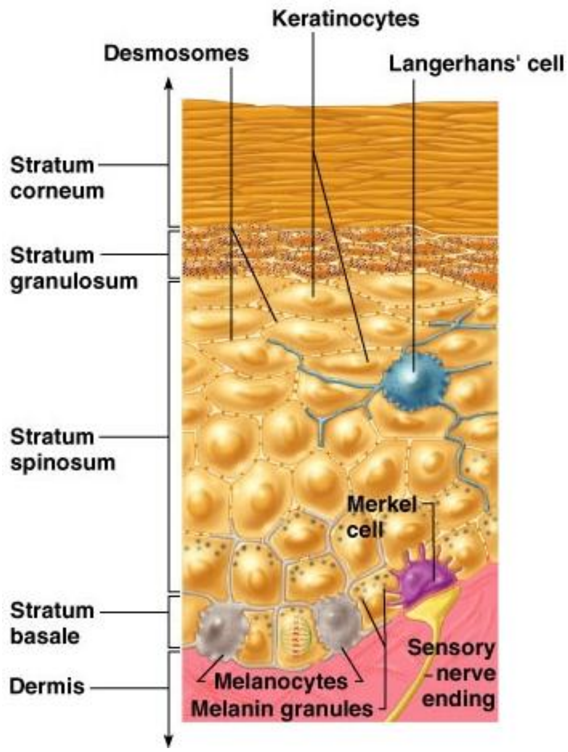
Colour of skin

- Depends on **several factors,** such as:
  3. Carotenoids which are yellow-orange-red pigments obtained from fruits and vegetables
  – Are found in cells of stratum corneum and fat in the adipocytes of hypodermis

  3. Amount of **oxygen saturated hemoglobin in the blood vessels of dermis**
  – Reduced oxygen saturated hemoglobin makes the skin appear paler, grayer or bluer

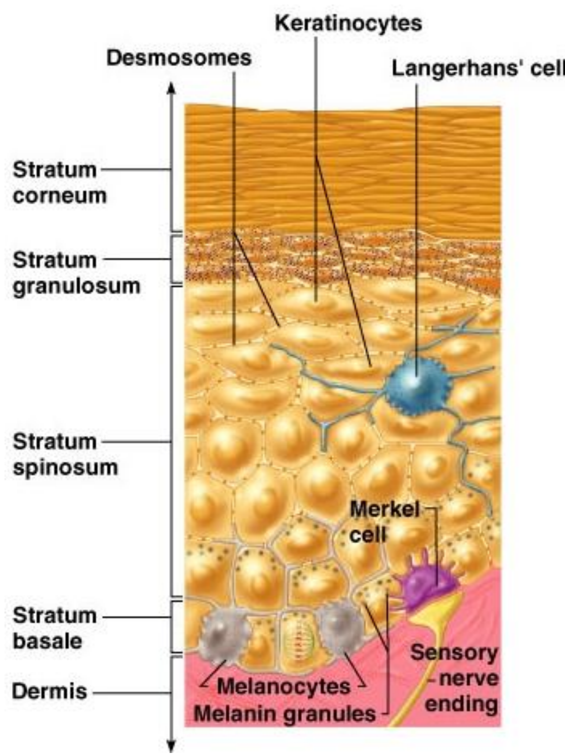
Langerhan’s cells

- **Star shaped cells** mainly in **the stratum spinosum**
- Belong to the **mononucleated phagocytic cells**



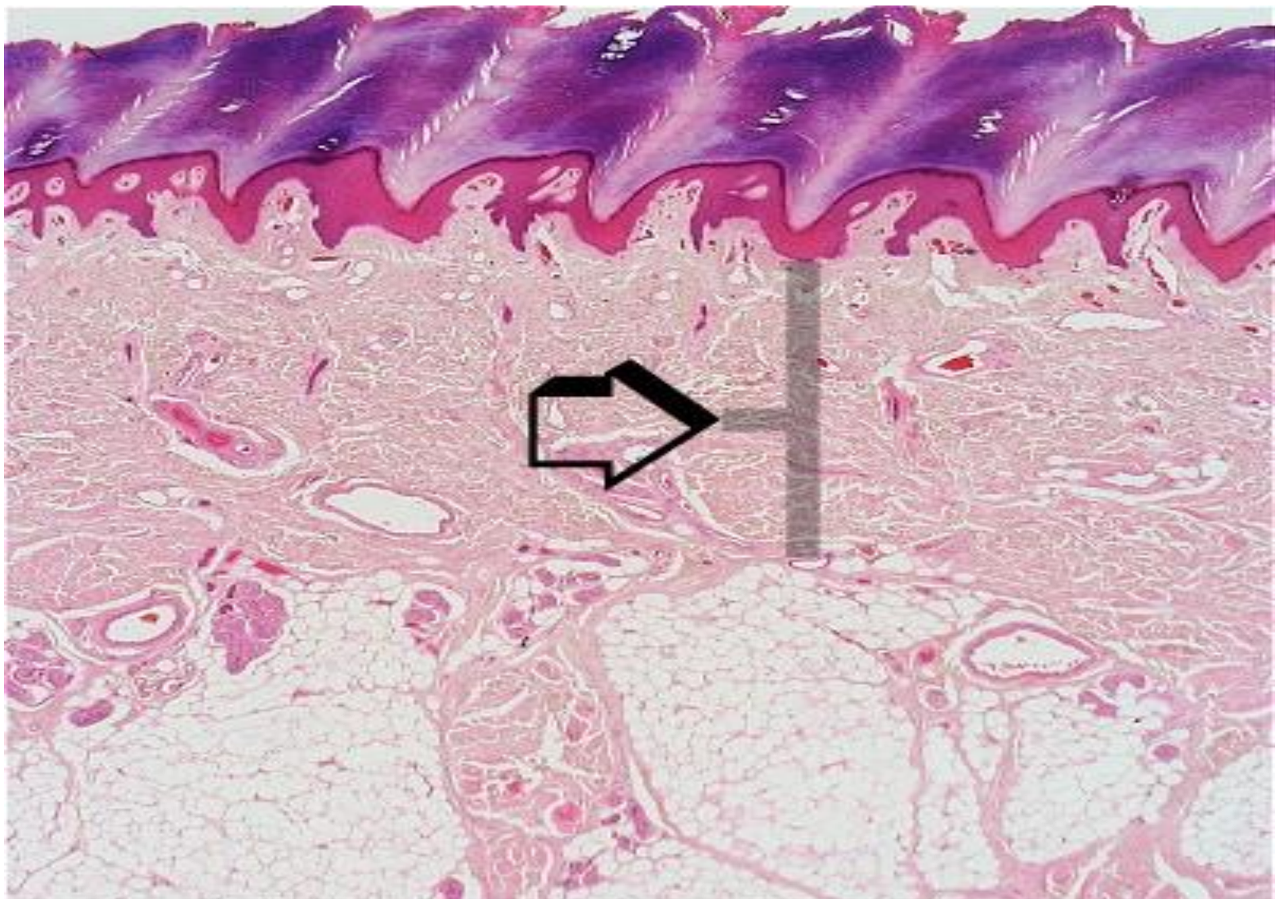
Merkel's cells

- Found **mainly in** the **thick skin of the palms** and **soles**
- Contacted by **free nerve endings** with expanded disk **(Merkel's disk)**
- May be **mechanoreceptors and/or neuroendocrine cells**



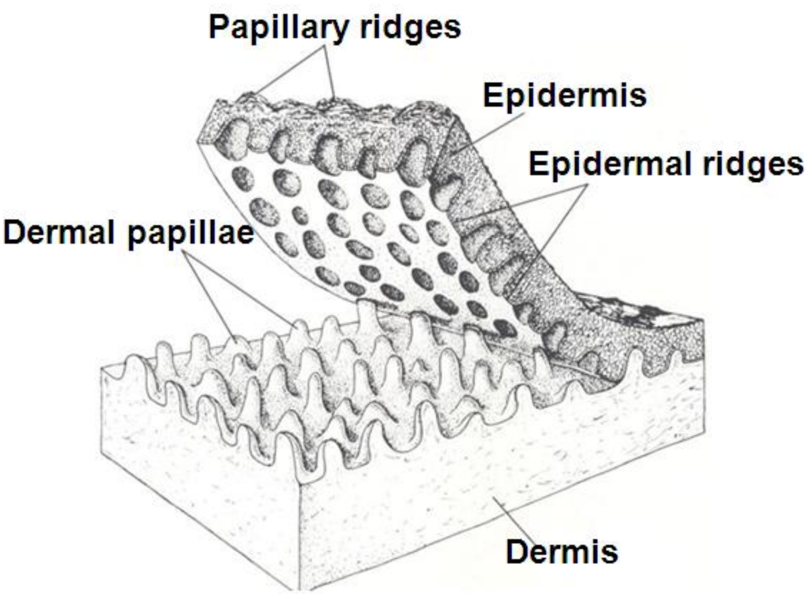
Dermis

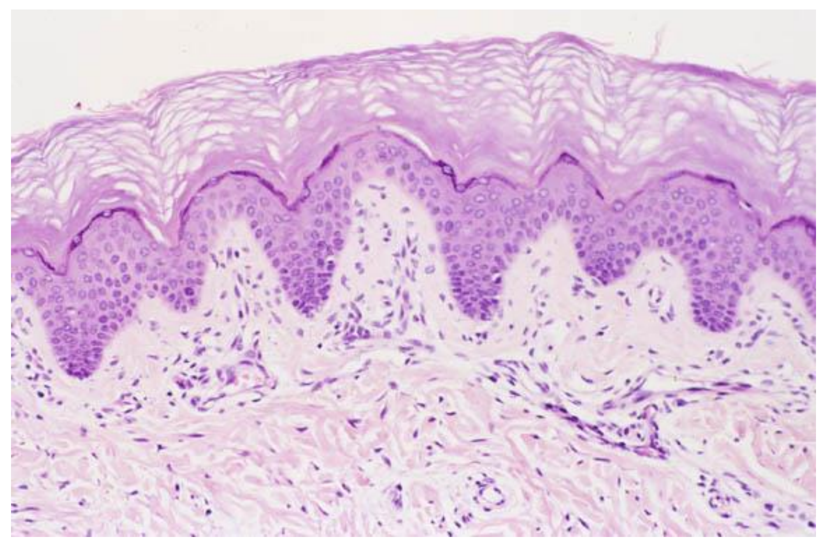
- Connective tissue layer



Dermis

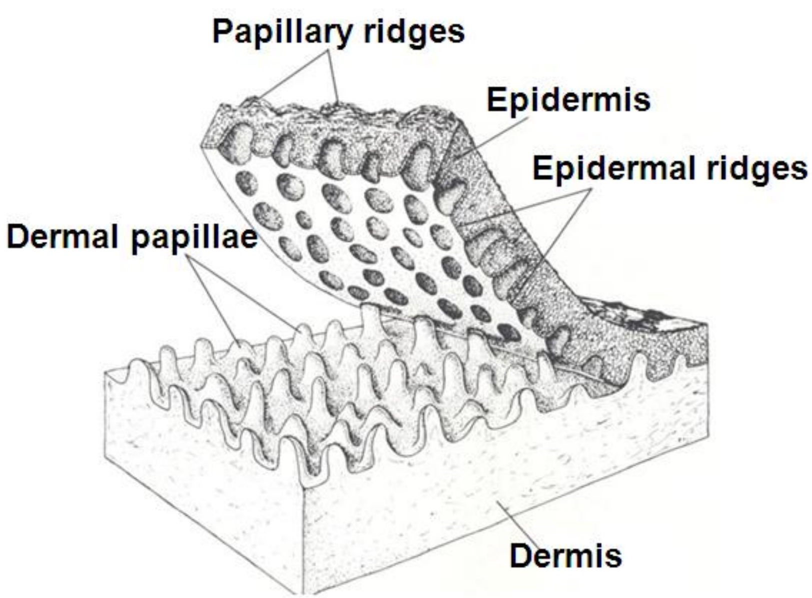
- Interlock with epidermis by **dermal papillae** and **epidermal ridges (pegs, or rete ridges)** for:
  - **Firm attachments**
  - Facilitating **diffusion** into the avascular epidermis

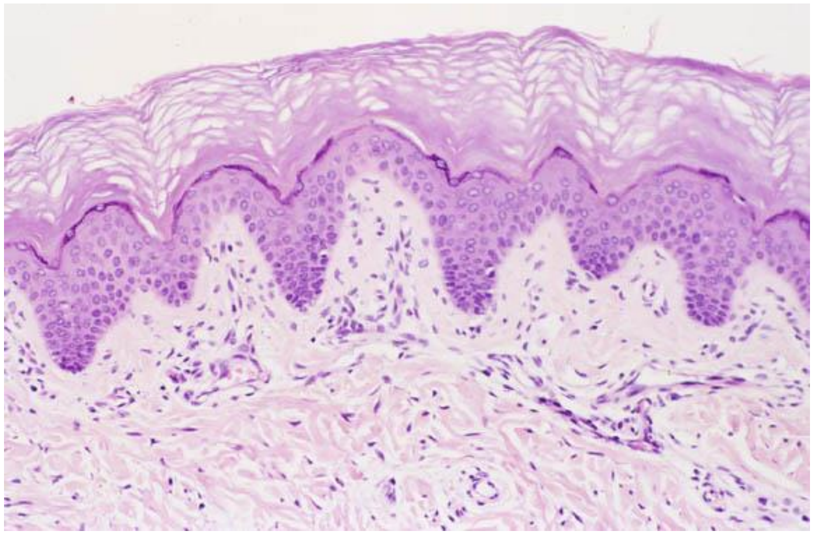




Dermis

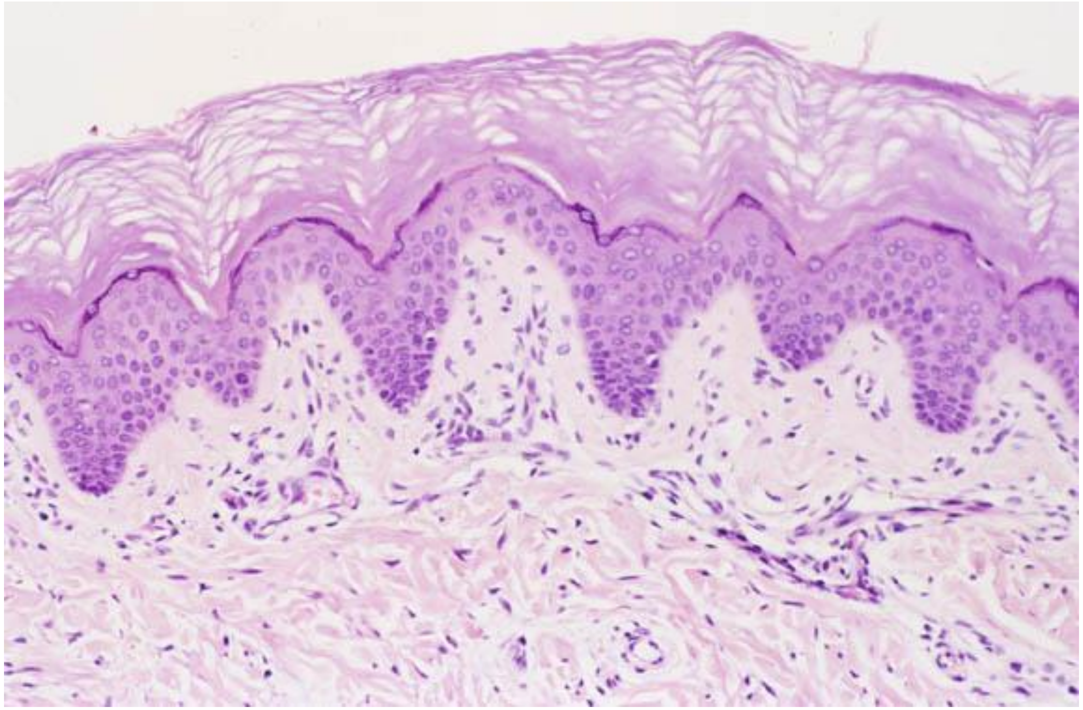
- In thick skin **dermal ridges** are also present
  - Have parallel arrangement with the dermal papillae located between them





Dermis

- Has two layers
  1. **Papillary layer**
  2. **Reticular layer**

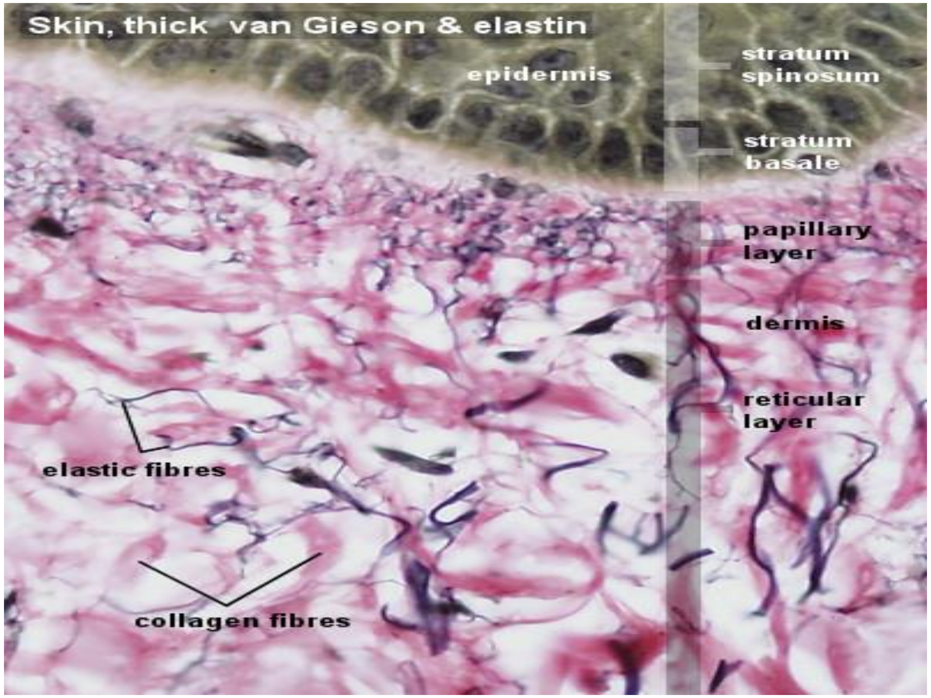


Dermis

1. **Papillary layer**

– Thin outer **loose connective tissue** layer

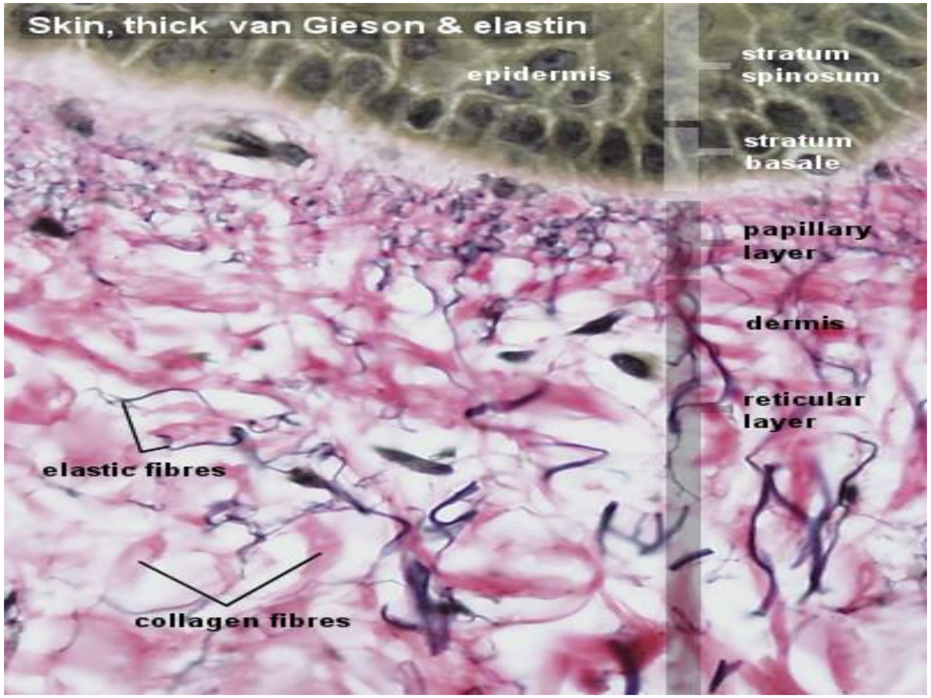
– Makes major part of dermal papillae



Dermis

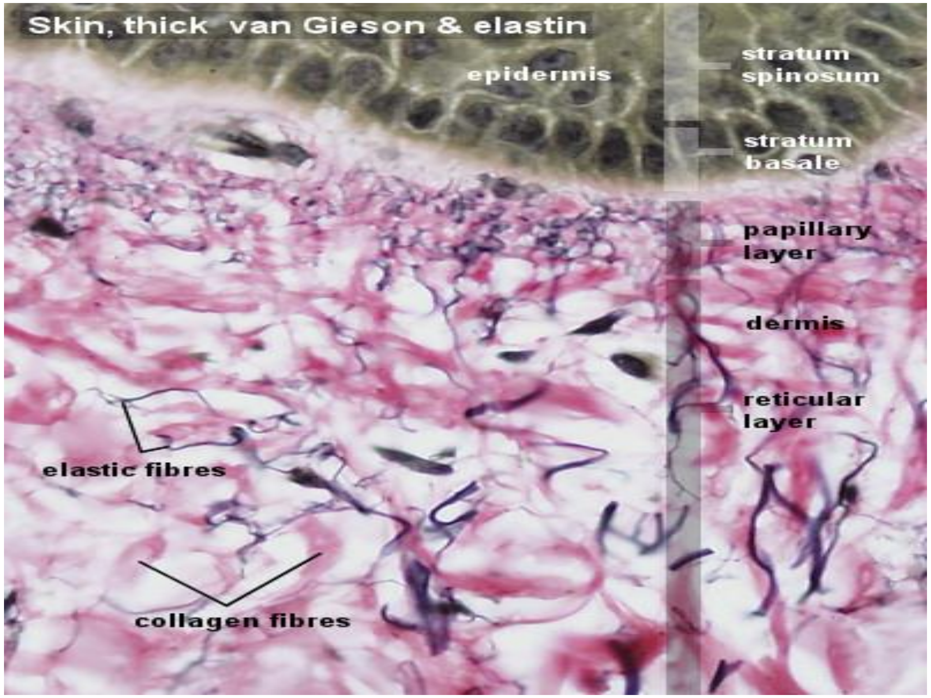
2. **Reticular layer**

- Thicker inner **dense irregular connective tissue layer**
- In the skin of **areola, penis, scrotum** and **perinium** contains a loose plexus of **smooth muscle** in its deeper aspect.



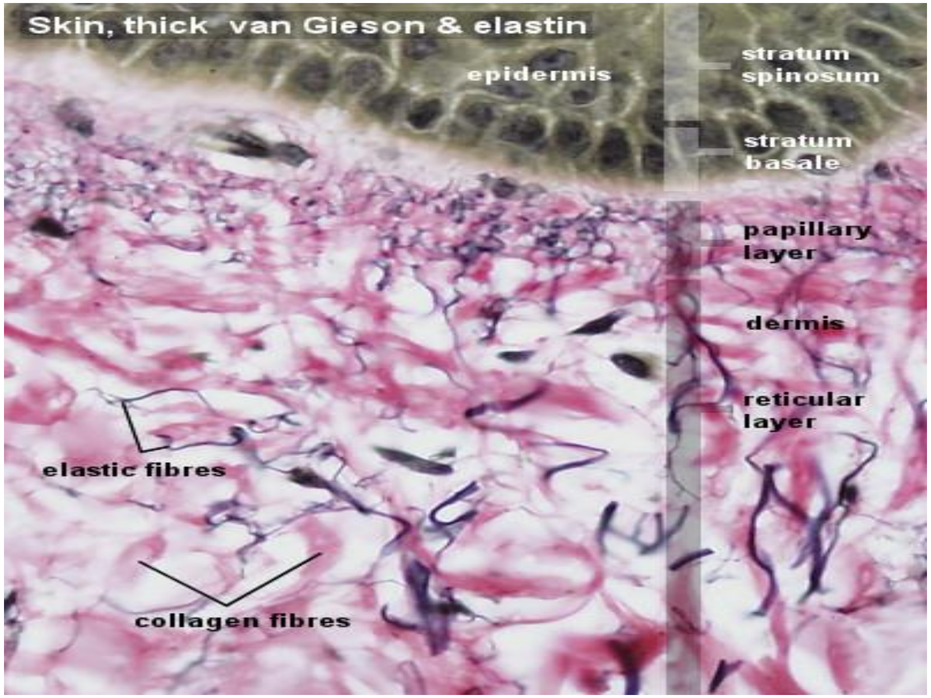
Dermis

- **With increasing age**
  - **Collagen** becomes **thicker** and **decrease** in **synthesis**
  - **Elastic** becomes **thicker** and **increase in number**



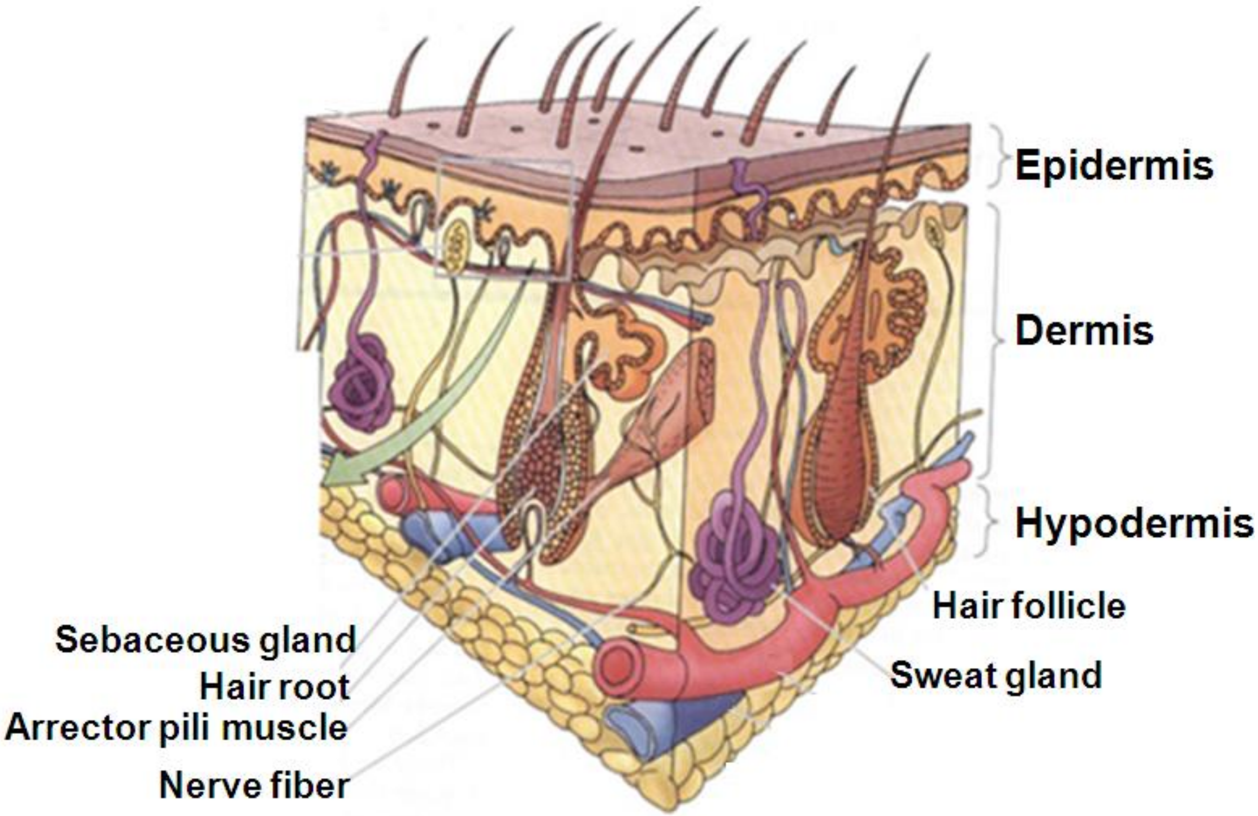
Dermis

- **In oldage**
  - **Collagen** will be **cross-linked**
  - **Elastic** fibres will be **lost** because of degeneration by sun.
  - As the result skin becomes **fragile, lose its elasticity** and **show wrinkles**



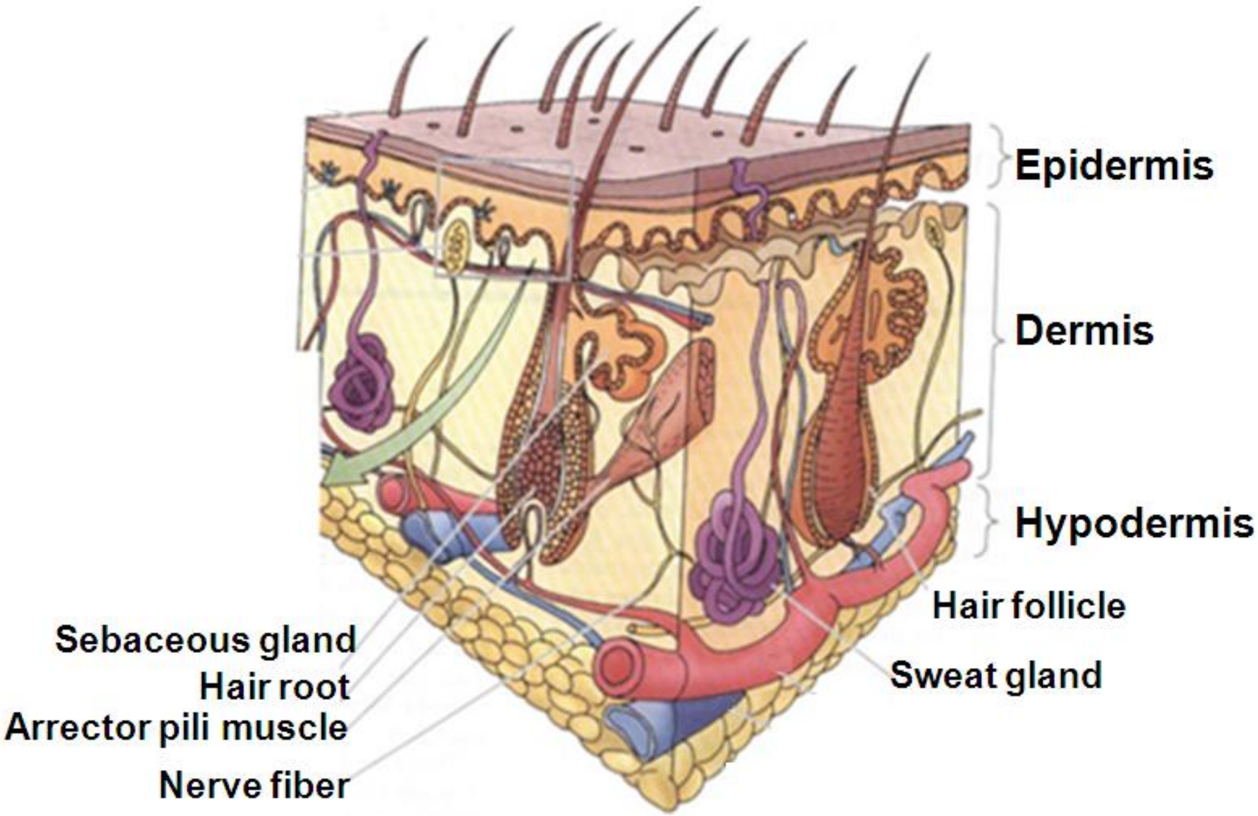
Dermis

- **Two arterial plexuses:**
  1. **Papillary plexus** - between papillary and reticular layers.
  2. **Cutaneous plexus** - between dermis and subcutaneous tissue.



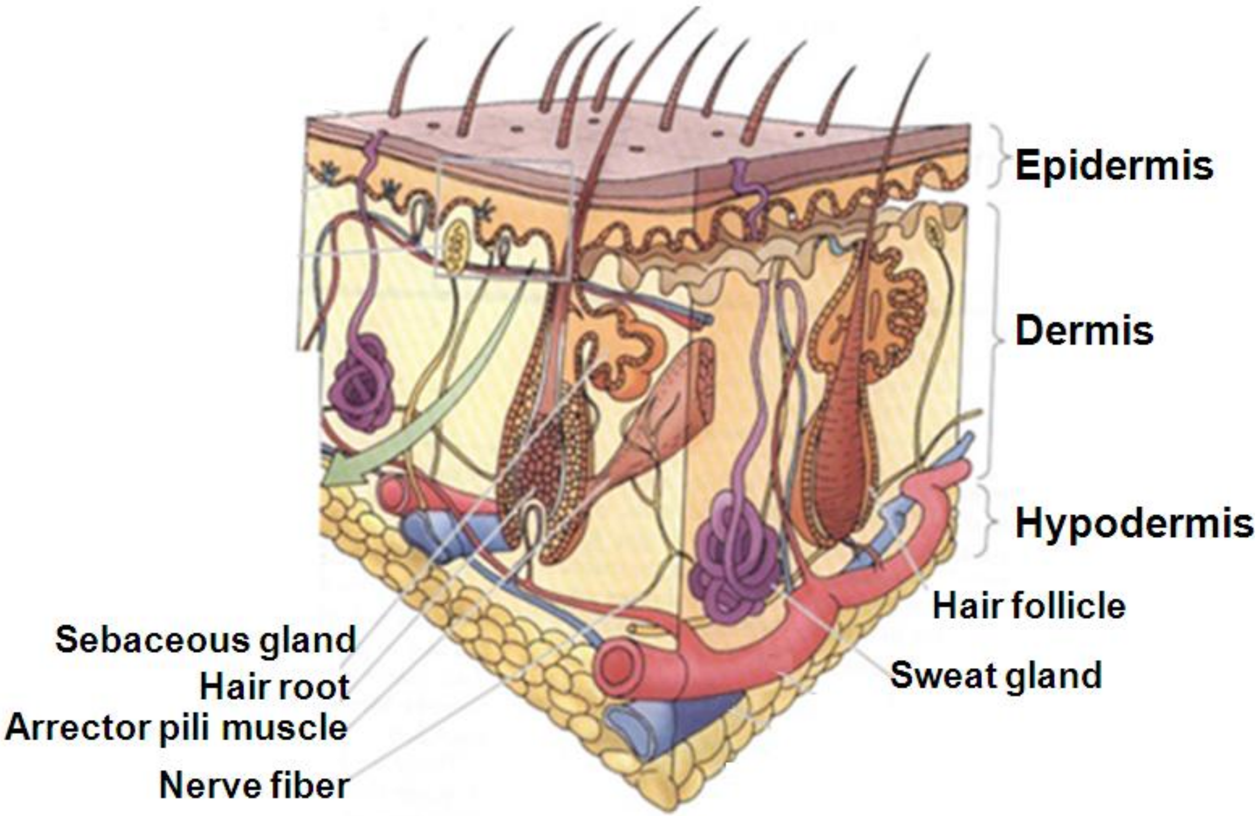
Dermis

- **Three venous plexuses** two found along the arterial plexuses and another one in between.
- **Arteriovenous anastomoses** allow shunting of blood **between papillary** and **cutaneous plexuses for temperature regulation**.



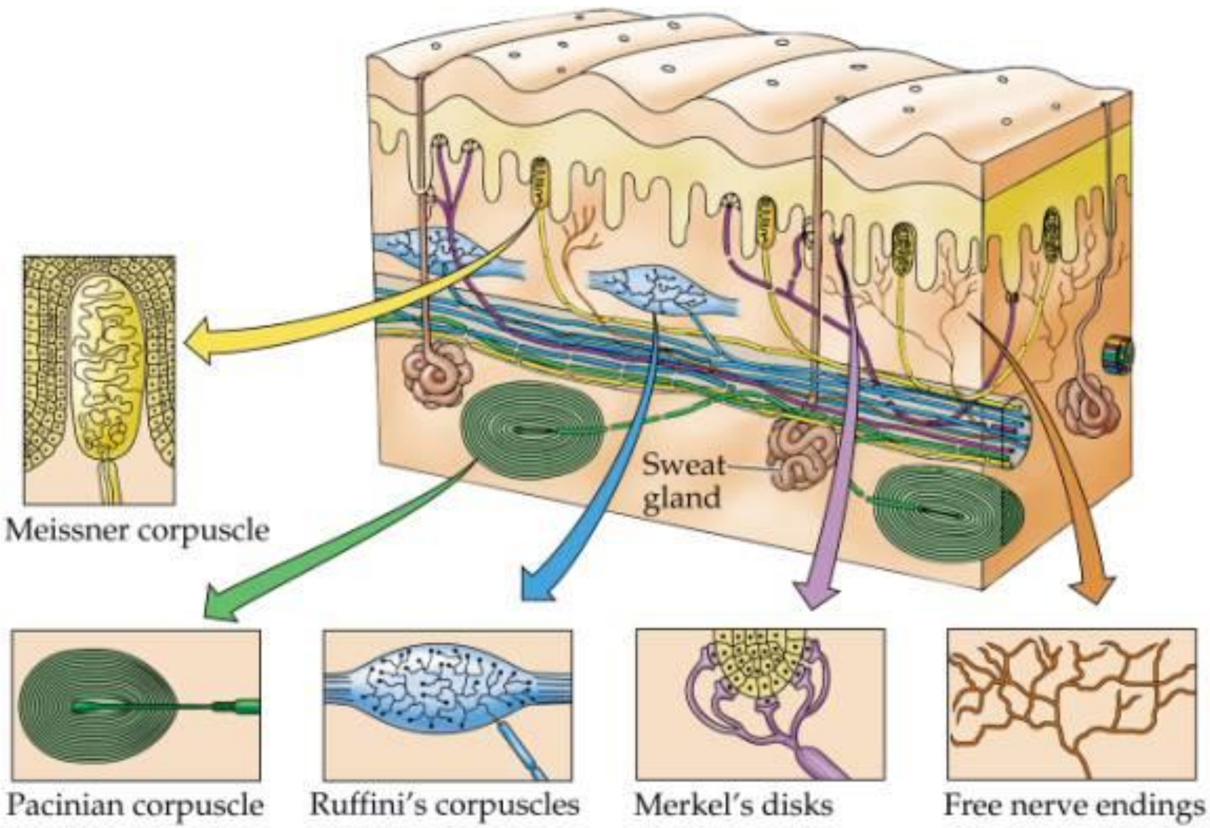
Dermis

- **Two lymphatic plexuses** found along the arterial plexuses



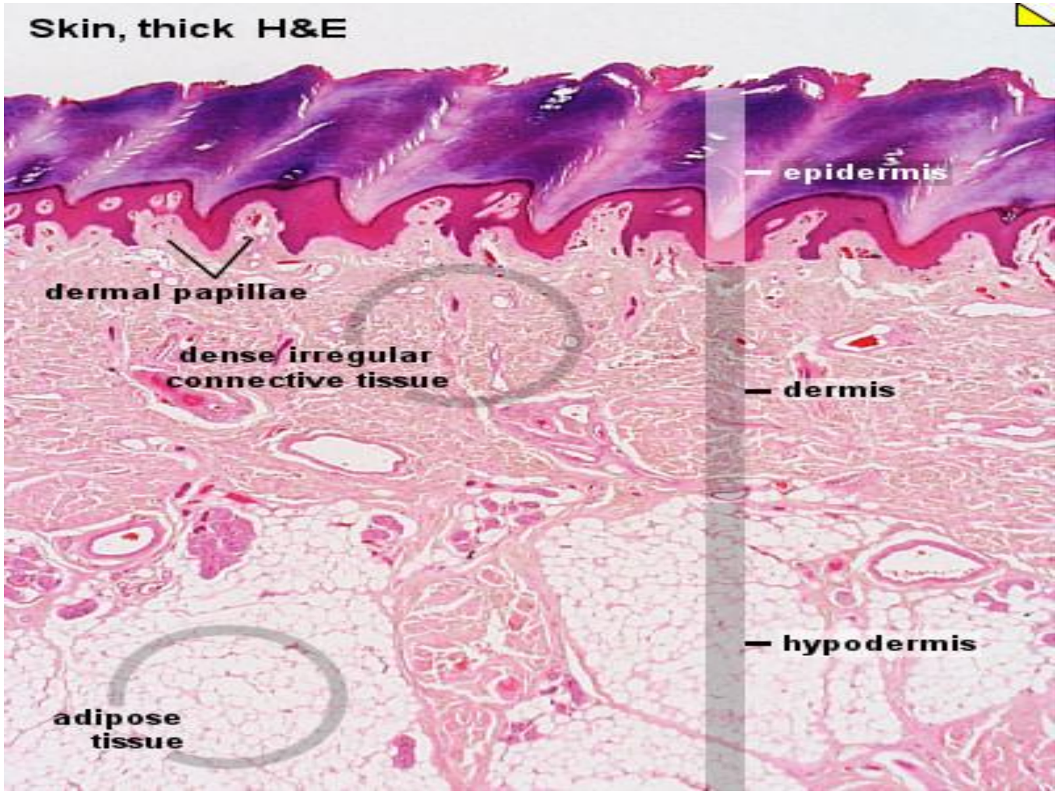
Dermis

- Rich **sensory innervation** made by different types of receptors, some penetrating into epidermis



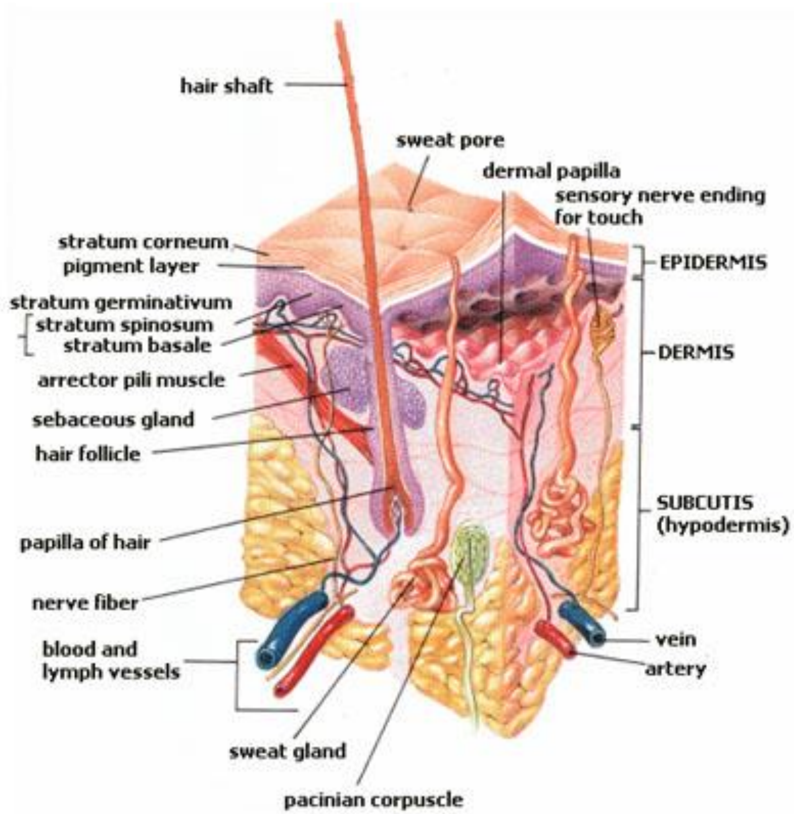
Subcutaneous tissue (hypodermis)

- Loose connective tissue with adipocytes
- Makes **superficial fascia**, known as **panniculus adiposus** when thick



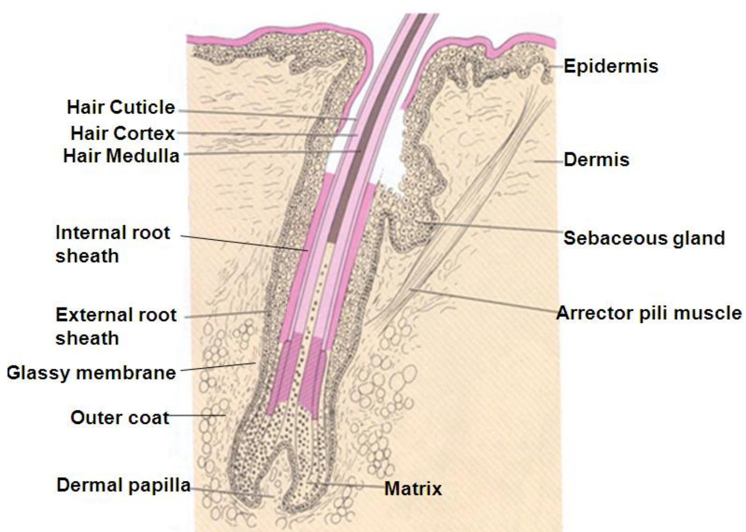
Hairs

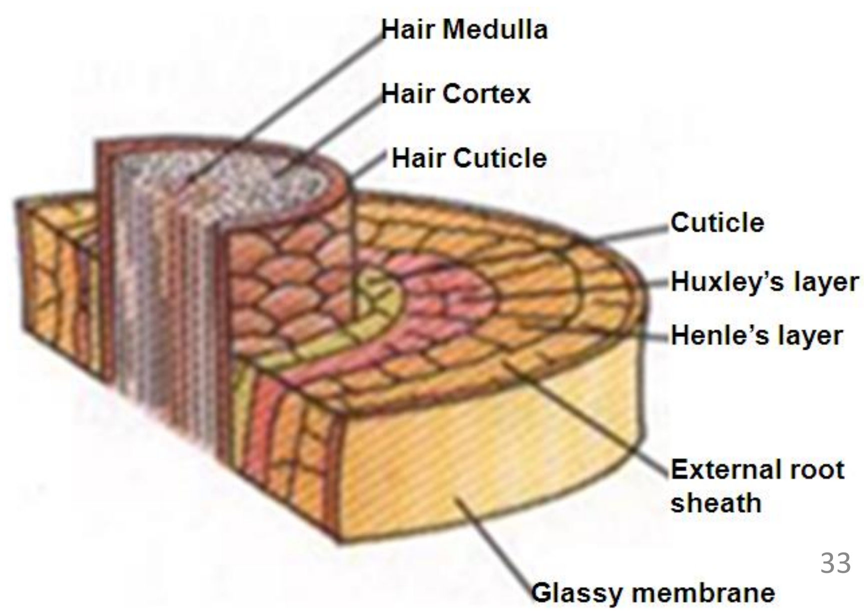
- By **dead** & **fused keratinized cells** of epidermis
- **No on palms, soles, lips, glans penis, clitoris and labia minora**
- Has two portions:
  1. **Root (hair follicle)**
  2. **Shaft**



Hair follicle

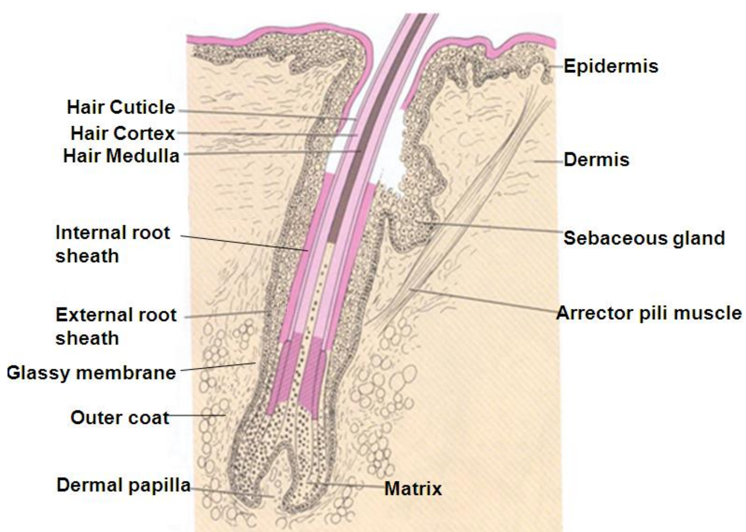
- End as **hair bulb** indented by **dermal papilla**
- **Core** is made by **cuticle, cortex** and **medulla** with two coats:
  1. **Outer coat** - by **condensation of dermis**
  2. **Inner coat (epithelial root sheath)**
  - Separated from outer coat by **glassy membrane** (modified basement membrane)

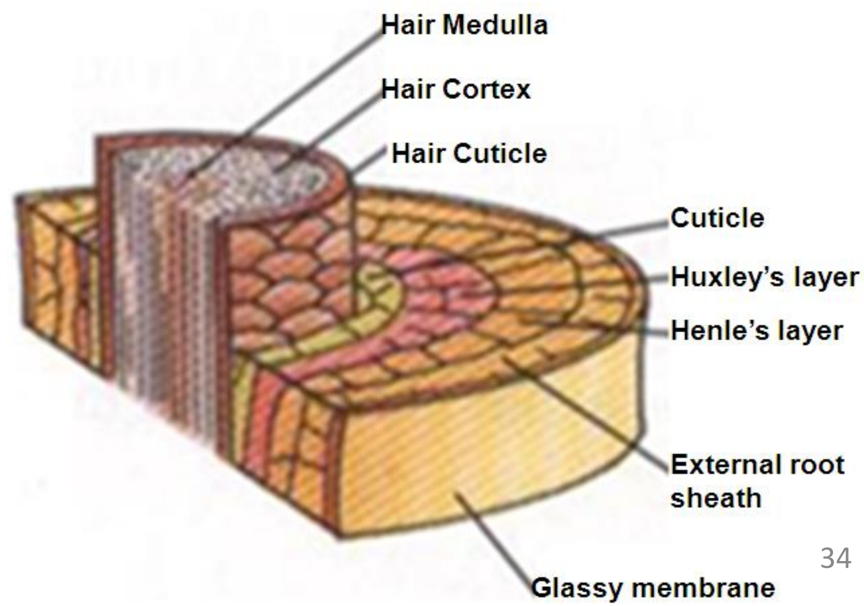




Hair Shaft

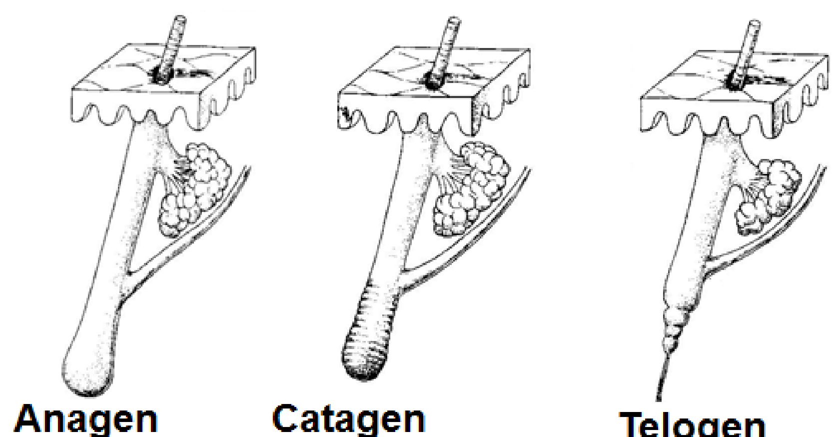
- Has three layers
  1. **Cuticle** – interlock with cuticle of inner epithelial sheath
  2. **Cortex** - make the **main part** of the hair
  3. **Medulla** - absent from fine hairs





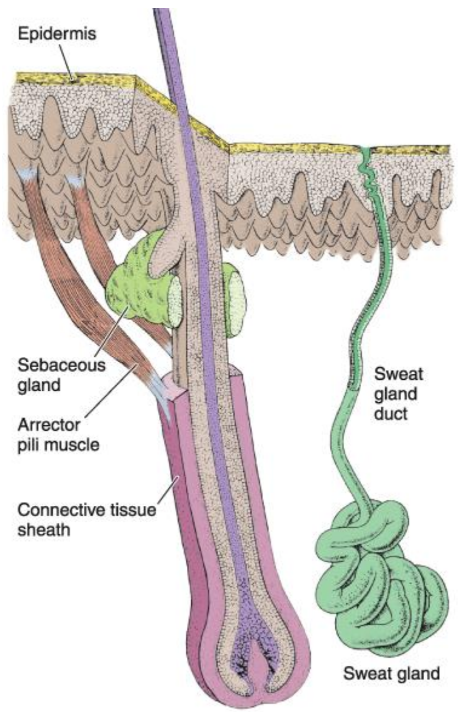
Life cycle of hairs in three phases

1. **Anagen phase**
   – A **growing period**
2. **Catagen pahse**
   – A short **involution phase**
3. **Telogen phase**
   – A **resting period** to be followed back by the **anagen pahse, or fall for shedding**



Arrector pili muscles

- Smooth muscle cells
- Run obliquely **from connective tissue around the hair follicle** to **papillary layer of dermis**
- Cause erection of hair shaft and goose flash

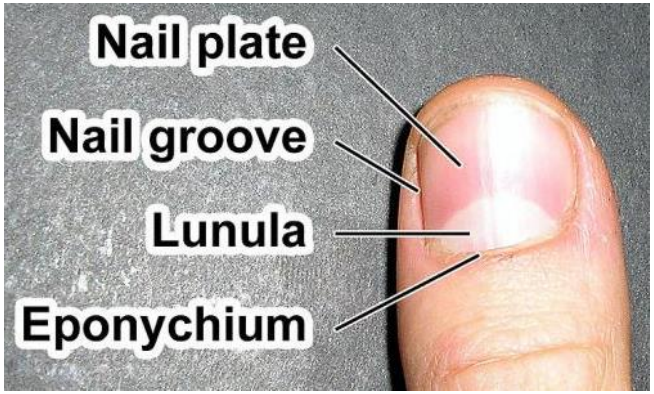


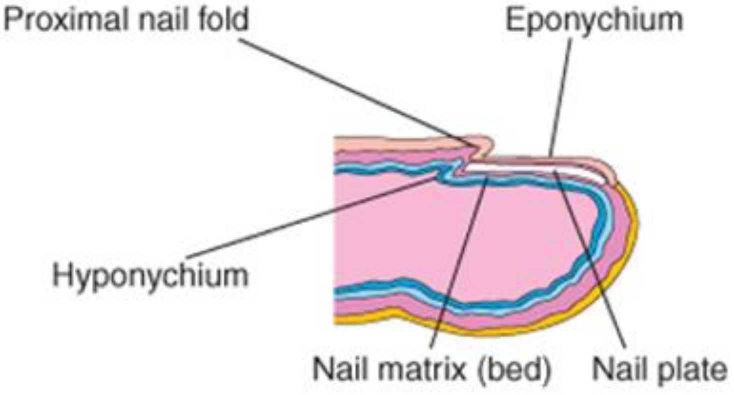
Graying of hair

- **Graying of hair** with increasing age is because of:
  - **Inability of melanocytes** to produce melanin
  - Increased appearance of **small air bubbles in the cortex** and **medulla**

Nails

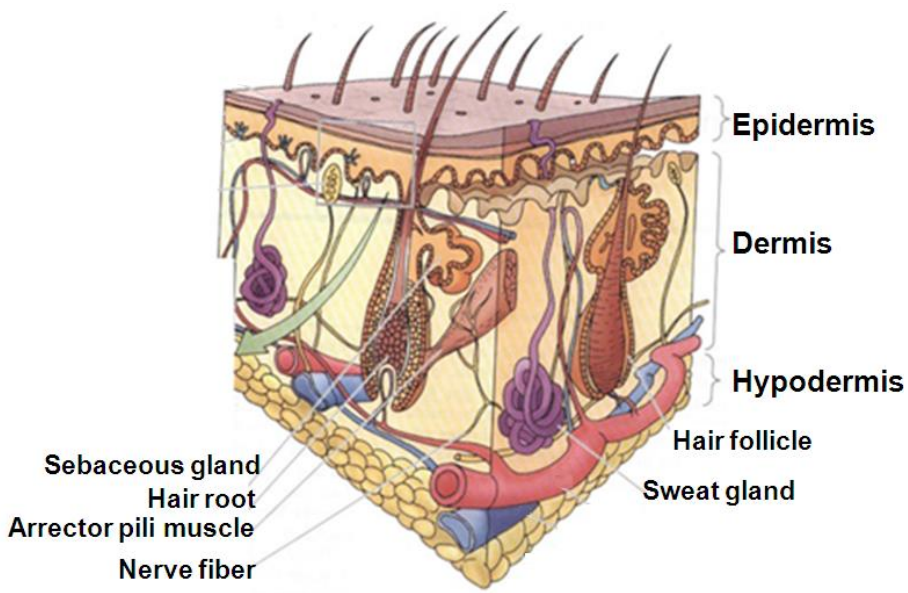
- Has the following **two regions**
  1. **Root**
  2. **Body (nail plate)**
     - Equivalent to the **stratum corneum** of epidermis
     - Rests on **nail matrix (bed) which:**
       – Is equivalent to **stratum germinativum** and **spinosum**
       – Has **lunule** a whitish area near the proximal nail folds made by **dividing cells** for growing of the nail





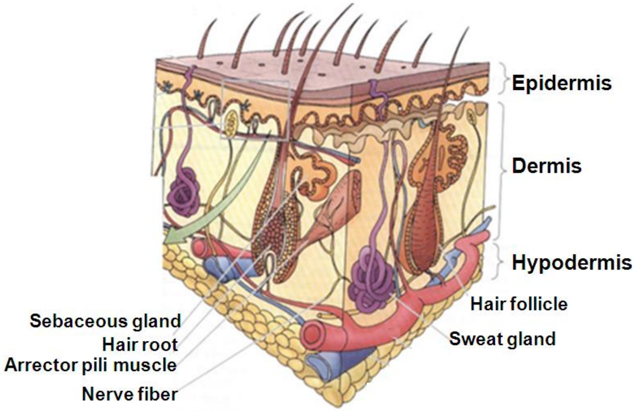
Glands of the Skin

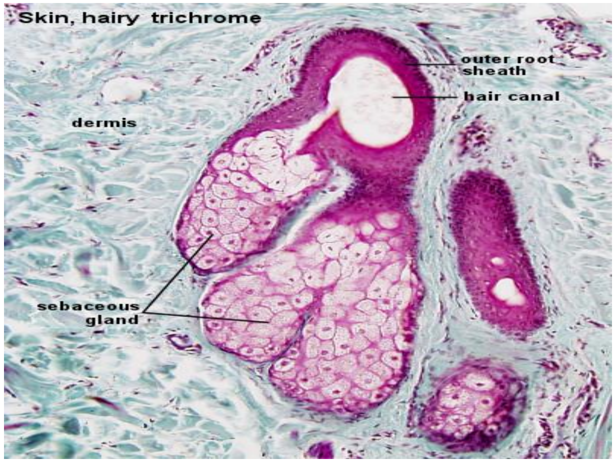
1. **Sebaceous gland**
2. **Sweat glands**
   a) **Eccerine (merocrine) sweat glands**
   b) **Apocrine sweat glands**



Sebaceous gland

- In the dermis of most region, **except on sole** and **palms**
- **Simple branched alveolar** - ducts open **onto upper portion of hair follicle** or **directly** onto the **surface of skin** in regions devoid of hairs
  - Is a **mixture of lipids**



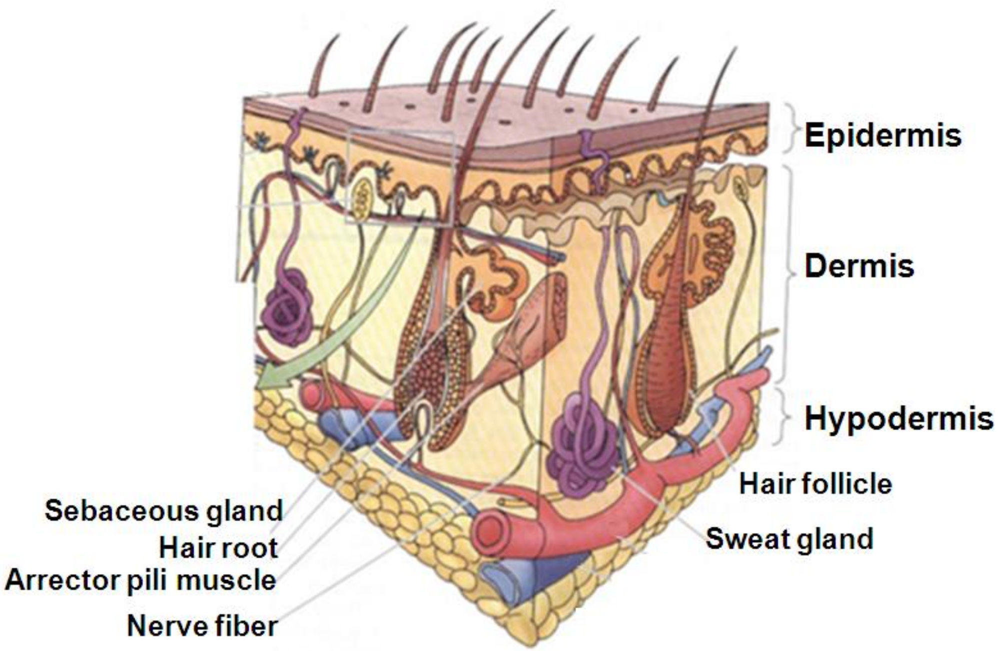


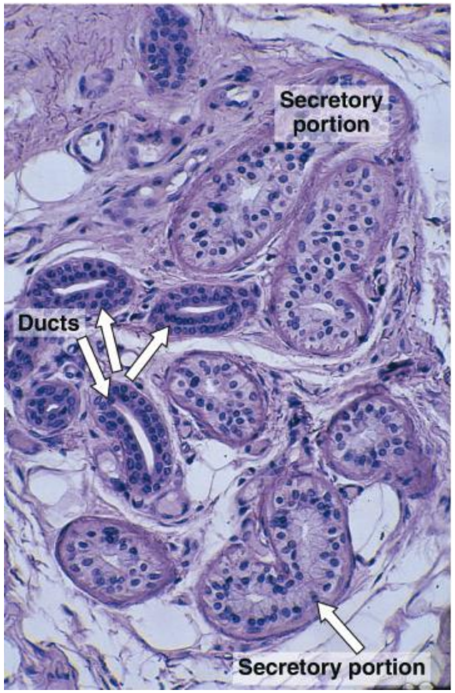
Sebaceous gland

- Produce **sebum**
  - Is a **mixture of lipids** with triglycerides, waxes, squalane, cholesterol and esters
  - **Continuously secreted** beginning **by puberty**, but if obstructed causes **acne**
  - Production is **controlled by testosterones and androgens**
  - **Functions** include
    - Lubrication of hairs
    - Oils surface of skin
    - Antibacterial and antifungal action

Eccerine (merocrine) sweat glands

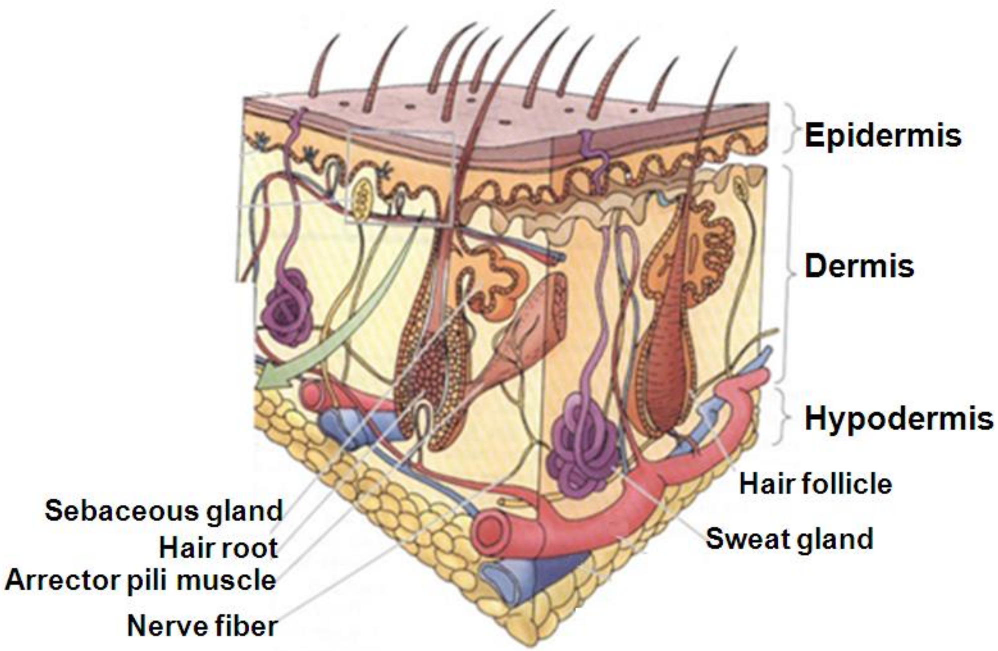
- Simple coiled tubular glands
- Secrete by merocrine mode

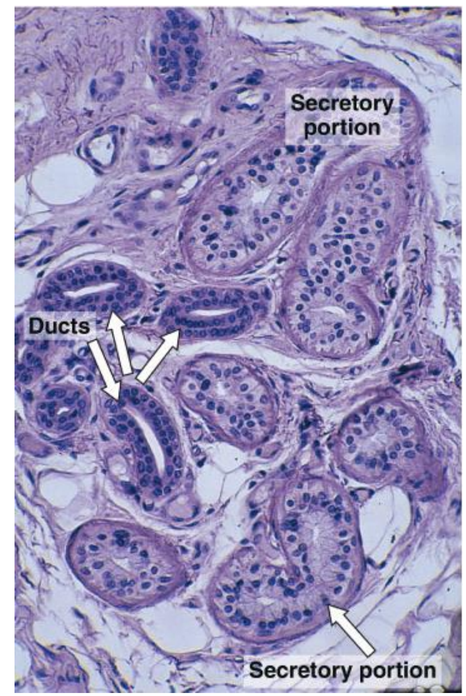




Eccerine (merocrine) sweat glands

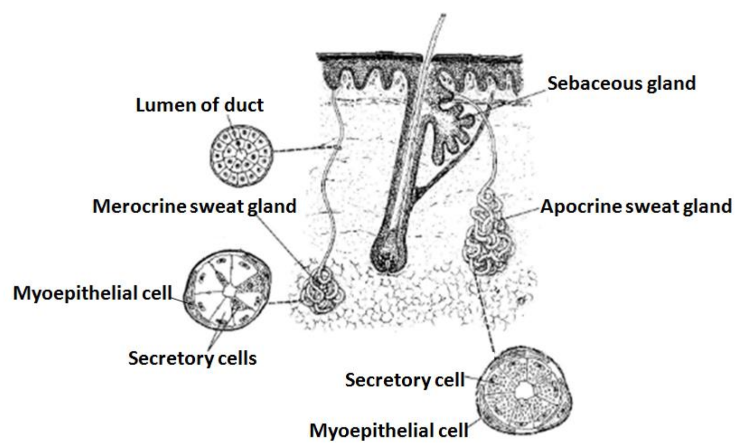
- Secrete a fluid secretion with H_2O, NaCl, urea, ammonia, uric acid and little protein that **evaporates causing cooling the skin surface**





Apocrine sweat glands

- **Larger simple coiled tubular glands**
- In axilla, nipple, anal region, scrotum, prepuce and labia minora
- **Secrete by merocrine mode**
- Open onto hair follicles above that of sebaceous glands
- Produce **viscous secretion**
- Modified apocrine sweat glands include:
  1. **Ciliary glands (glands of Moll)** in the margin of the **eyelids**
  2. **Ceruminous glands** in the **external acoustic meatus**
  3. **Circumanal glands** in the anal canal




Thank
you